

M5: Advanced Android App Architecture and Development Practices

June 9, 2026 4:22 PM

Overview

Introduction

Hello and welcome to Module 5 of COMP-304. In this module, you will learn about Advanced Android App Architecture and Development Practices.

Learning Outcomes

At the end of this module, you will be able to:

1. Discuss the fundamental principles of app architecture and its importance in Android development.
2. Implement the Model-View-View Model(MVVM) architecture to enhance code separation and maintainability.
3. Explain the functionality and benefits of Lazy Column in Jetpack Compose.
4. Create composable functions using Lazy Column to efficiently display lists.
5. Utilize Jetpack libraries to streamline and enhance app development.
6. Apply dependency injection to improve code modularity and testability.
7. Migrate an Android project to Kotlin Gradle DSL and manage dependencies using version catalogs.

Checklist

There are several tasks you'll need to complete:

1. Review the online content for this module, which includes reviewing the module's key terms and concepts.
2. Complete Lab Session Week 5.

Key Terms and Concepts

Here are a few terms you need to be familiar with.

Term	Definition
App Architecture	The framework that organizes code to support scalability, maintenance, and development efficiency.
MVVM (Model-View-ViewModel)	An architecture pattern recommended by Google that separates the data model layer from the user interface via the ViewModel.
Model Layer	Manages the data and business logic in the MVVM architecture.
View Layer	Handles the layout and display in the MVVM architecture.
ViewModel Layer	Acts as a bridge between the Model and the View, managing data flow and providing an abstraction layer to the UI.
Dependency Injection	A design pattern used to improve code modularity and testability by decoupling object creation from their usage.
Kotlin Gradle DSL	A domain-specific language for defining Gradle build scripts in Kotlin, offering code autocompletion, type safety, and compile-time error detection
Version Catalogs	A feature in Gradle for central management of dependencies, simplifying updates and ensuring consistency across modules.

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Weekly Topic

The topic this week is about **Advanced Android App Architecture and Development Practices**.

The following resources provide great insights into **Android App Architecture** :

1. <https://developer.android.com/topic/architecture>
2. [Android MVVM](#)
3. [Dependency injection in Android](#)

Summary

Key Take-Aways

Here are a few key take-aways from this module.

- Implementing MVVM architecture
- Using Jetpack libraries
- Implementing dependency injection
- Using Kotlin Gradle DSL
- Using version catalogs.

Summary Checklist

Don't forget to complete the items on your checklist:

1. Review the online content for this module, which includes reviewing the module's key terms and concepts.
2. Read articles/tutorials given in Learning Activities.
3. Work on Lab assignment 2.

Works Cited

1. <https://developer.android.com/topic/architecture>

Coming up next, we'll be looking at the weekly topic for module 6 as identified in the approved course outline.

COMP-304: Mobile Apps Development